MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 5

1. *Determine which of the following sets are subspaces of the indicated vector spaces and* **explain why**. *Also sketch the sets W_3 and W_4 and give geometric reasons why W_3 and W_4 either are or are not subspaces of $\mathbb{R}^2$.*

(a) $W_1 = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \ : \ a - 2b = c, 4a + 2c = 0 \right\} \subseteq \mathbb{R}^3.$

(b) $W_2 = \left\{ \begin{bmatrix} a - b \\ c \\ a + c \\ a - 2b - c \end{bmatrix} \ : \ a, b, c \in \mathbb{R} \right\} \subseteq \mathbb{R}^4.$

(c) $W_3 = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \ : \ a \cdot b \geq 0 \right\} \subseteq \mathbb{R}^2.$

(d) $W_4 = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} \ : \ b \geq 0 \right\} \subseteq \mathbb{R}^2.$

(e) $W_5 = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \ : \ a - 2b = 1, 4a + 2c = 0 \right\} \subseteq \mathbb{R}^3.$

2. *Are the following sets subspaces of the vector space of all 2×2 matrices $M_{2\times 2}$?*

(a) $\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{2\times 2} \ : \ ad - bc = 0 \}$.

(b) $\{B \in M_{2\times 2} \ : \ B^T = B\}$. (This is the set of symmetric 2×2 matrices.)

3. *Let $\mathbb{P}_n$ be the vector space of polynomials of degree at most n. Let W_1 be the set of all polynomials $p(t)$ in $\mathbb{P}_n$ which have a zero at $t = 1$ (that is, $p(1) = 0$), and let W_0 be the set of all polynomials $p(t)$ in $\mathbb{P}_n$ which have a zero at $t = 0$. Are these sets subspaces of the vector space of all polynomials?*

4. *In the class we introduced the vector space V of functions from $\mathbb{R}$ to $\mathbb{R}$. This problem will discuss a few aspect of this vector space.*

(a) Observe that $\sin(x), \cos(x)$ are in V and so are $2\sin(x)$ and $\sin(2x)$. Are $2\sin(x)$ and $\sin(2x)$ linear combinations of $\sin(x)$ and $\cos(x)$?

(b) Is the set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of V?

(c) Is the set of all continuously differentiable functions from $\mathbb{R}$ to $\mathbb{R}$ a subspace of V?

5. *Find an explicit description of the nullspaces of the following two matrices. You should write the nullspaces as spans of sets of vectors.*

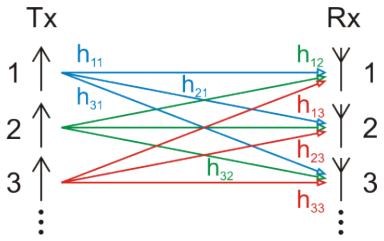
$$A = \begin{bmatrix} 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

6. *Construct a matrix A such that*

$$\mathrm{Nul}(A) = \mathrm{span}\left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right).$$

7. *Let A, B be two matrices such that the product AB is defined. Is the nullspace of A contained in the nullspace of AB? Is the nullspace of B contained in the nullspace of AB? Explain why or why not.*

8 (Wireless communication)**.** *The goal of this exercise is to better understand what a null space and a column space of a matrix represent in practical applications. In multiple-input and multiple-output (short: MIMO) systems, a transmitter sends multiple streams by multiple transmit antennas.*



Suppose there are n transmitters and m receivers. This can modeled using Linear Algebra:

$$\underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}}_{\text{received vector } \mathbf{y}} = \underbrace{\begin{bmatrix} h_{1,1} & \dots & h_{1,n} \\ \vdots & \ddots & \vdots \\ h_{m,1} & \dots & h_{m,n} \end{bmatrix}}_{\text{channel matrix } H} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{\text{transmitted vector } \mathbf{x}}.$$

So here the $\mathbf{x}$ vector describes what the transmitter is sending out, and $\mathbf{y}$ is the vector describing what is received. The channel matrix H tells you how the signal $\mathbf{x}$ is transformed in to the vector $\mathbf{y}$.

(a) In terms of what is received by the receiver, what does the first column of H represent? What does the i-th column represent?

(b) If the signal $\mathbf{x}$ belongs to the nullspace of H, what signal $\mathbf{y}$ will be received?

(c) In a well designed system, what you want $\mathrm{Nul}(H)$ to be?

(d) In terms of transmitted signals, what is $\mathrm{Col}(H)$?

(e) If you take away one of the transmitters, how does H change and how does the column space of H change? What does that mean in terms of signals you can transmit?

9 (Redox reactions)**.** *Balance the following redox reaction:*

$$x_1\mathrm{H}^+ + x_2\mathrm{Cr_2O_7^{2-}} + x_3\mathrm{C_2H_5OH} \leftrightarrow x_4\mathrm{Cr^{3+}} + x_5\mathrm{CO_2} + x_6\mathrm{H_2O}.$$

Describe the subspace of vectors $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}$ that balance this equation. (How can you make sure that the subspace is spanned by a single vector? Look at the electric charge!)